Identification Note: Demand Shifters and Price Elasticity in Asset Markets

Goal

Identify the *price elasticity of investor demand* for a security: how much the market price must move to absorb an exogenous change in holdings demanded by constrained investors, holding cash-flow fundamentals fixed.

Key identification problem

Observed (P, Q) co-move because *both* cash-flow news (fundamentals) and investor demand (constraints, mandates) move at the same time. Without an exogenous shifter, the slope of demand is not identified from raw correlations.

Identification idea (IO analogy)

Borrow the IO logic (Bresnahan-style): use an *exogenous demand shifter* Z that shifts investor demand but does not directly change expected cash flows. Then $(\Delta Q, \Delta P)$ around Z traces the demand curve.

Minimal setup

Let security i clear in the market:

$$Q_{i,t} = D_{i,t}(P_{i,t}, X_{i,t}; Z_{i,t}), \qquad \frac{\partial D}{\partial P} < 0,$$

where $X_{i,t}$ are fundamentals (cash-flow/risk characteristics) and $Z_{i,t}$ is a demand shifter (regulation/index/mandate).

Linearized around the event window:

$$\Delta Q_i \approx \beta\, \Delta P_i + \gamma\, \Delta Z_i + \text{(small residual)}.$$

If ΔZ_i is exogenous and affects demand but not fundamentals, then instrumenting ΔP_i with ΔZ_i

identifies β:

$$\widehat{\beta}^{IV} = \frac{\mathrm{Cov}(\Delta Q_i, \Delta Z_i)}{\mathrm{Cov}(\Delta P_i, \Delta Z_i)}.$$

Elasticity (local) can be reported as

$$\widehat{\varepsilon} = \widehat{\beta}^{IV} \cdot \frac{\bar{P}}{\bar{Q}} \quad \text{(sign negative under downward-sloping demand).}$$

Interpretation

Large $|\Delta P|$ for a given forced ΔQ implies *inelastic* demand (limited risk-bearing capacity / strong constraints). Small $|\Delta P|$ for large flows implies *elastic* demand.

Candidate demand shifters Z

- Regulatory capital or risk-weight changes (RBC buckets, leverage constraints).
- Index inclusion / benchmark reconstitution and mechanical tracking flows.
- Mandate/accounting frictions (HTM vs AFS, ESG screens, duration targeting).

Exclusion restriction and diagnostics

- **Exclusion**: Z should not shift expected cash flows in the event window.
- Placebos: pre-trends, placebo dates, unaffected securities/holders.
- Cross-sectional heterogeneity: strongest effects where constrained holders are dominant.

Figure